

PES University - Department of Computer Science & Engineering

Lab 1: Requirements Engineering & UML Use-Case Modelling

Problem Statement #55: Book Club Reading Challenge & Discussion Portal

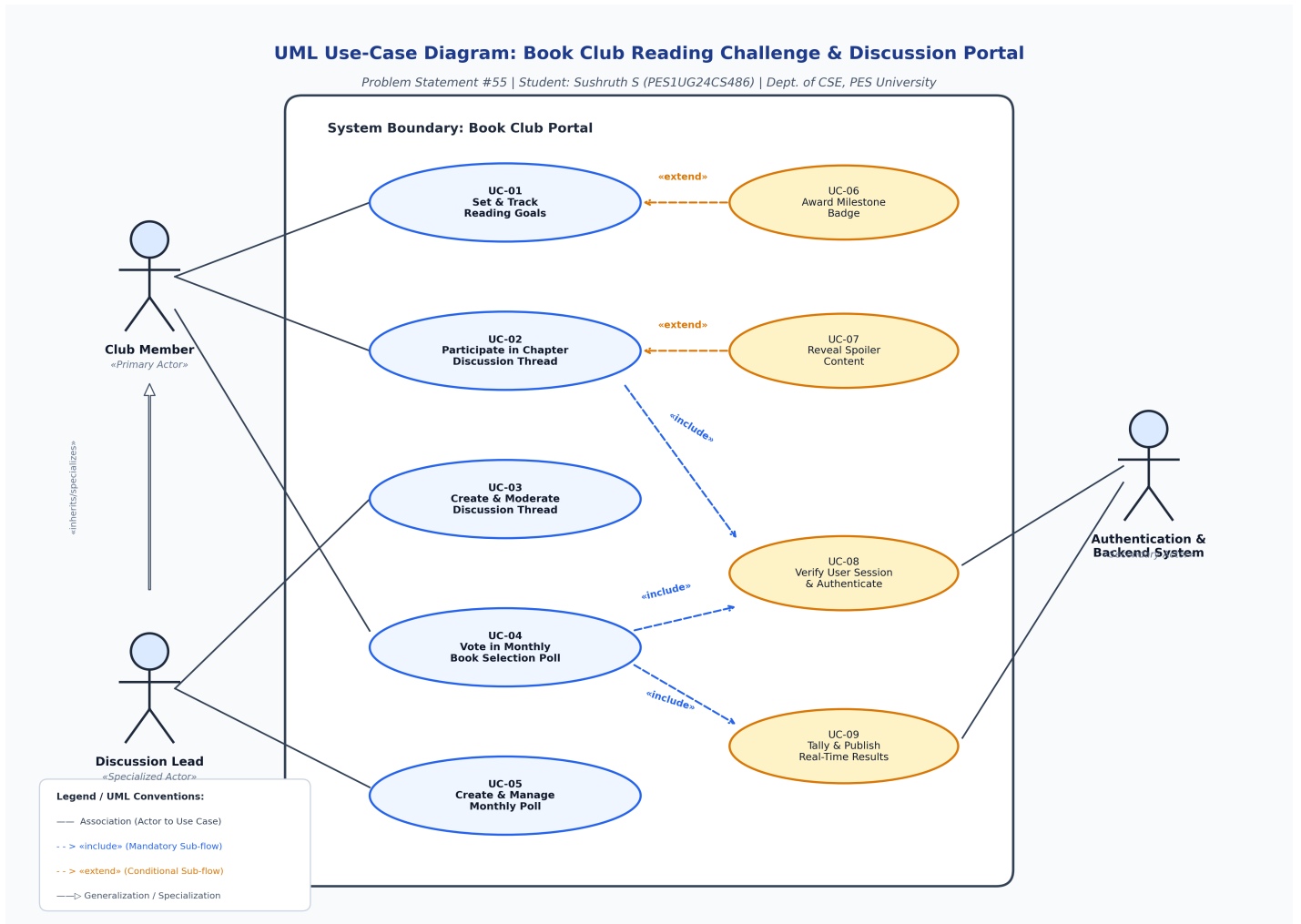
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1. Complete Requirements Specification Table (5 FRs, 2 NFRs)

| Req ID  | Type                   | Description                                                                                                                                       | Priority | Acceptance Criteria                                                                             | Rationale                                        | Comments                                 |
|---------|------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|----------|-------------------------------------------------------------------------------------------------|--------------------------------------------------|------------------------------------------|
| FR-001  | Functional             | The system shall hide content inside spoiler-tagged discussion comments until the user explicitly clicks to reveal or marks that chapter as read. | High     | <b>Pass:</b> Masked with blur filter.<br><b>Fail:</b> Exposed in search preview.                | Protects readers from inadvertent plot spoilers. | Client CSS mask + read-state sync.       |
| FR-002  | Functional             | The system shall track personal reading page goals and update progress toward active challenge.                                                   | High     | <b>Pass:</b> Gauge increments with valid pages.<br><b>Fail:</b> Negative page count accepted.   | Gamifies community reading targets.              | Boundary validation against total pages. |
| FR-003  | Functional             | The system shall allow Discussion Leads to create and moderate chapter-specific discussion threads.                                               | High     | <b>Pass:</b> Leads can pin/close threads.<br><b>Fail:</b> Regular members access lead actions.  | Ensures structured, safe book analysis.          | RBAC on moderation endpoints.            |
| FR-004  | Functional             | The system shall allow Club Members to cast exactly one vote in the active monthly book selection poll.                                           | High     | <b>Pass:</b> 1 vote recorded, buttons lock.<br><b>Fail:</b> Duplicate vote on refresh.          | Democratic, tamper-proof book choice.            | Compound index (UserID+PollID).          |
| FR-005  | Functional             | The system shall award digital badges upon reaching reading milestone goals (e.g. 500 pages).                                                     | Medium   | <b>Pass:</b> Badge shows within 5s of milestone.<br><b>Fail:</b> Badge granted without goal.    | Increases engagement & retention.                | Asynchronous milestone trigger.          |
| NFR-001 | Non-Func (Perf & Sec)  | Monthly poll voting must prevent duplicate votes with session verification & real-time tallies.                                                   | High     | <b>Pass:</b> Latency < 200ms at 1,000 peak users.<br><b>Fail:</b> Duplicate vote or lag > 1.5s. | Guarantees poll integrity & speed.               | Redis cache + atomic DB transaction.     |
| NFR-002 | Non-Func (Sec & Avail) | System shall maintain 99.9% uptime and encrypt credentials and history using AES-256 / SHA-256.                                                   | High     | <b>Pass:</b> Uptime ≥ 99.9%, zero PII leak.<br><b>Fail:</b> Monthly downtime > 43.8 min.        | Ensures privacy and high reliability.            | TLS 1.3 + column encryption at rest.     |

## 2. UML Use-Case Diagram

The diagram models all core actors (Club Member, Discussion Lead, Authentication & Backend System) and primary use cases (UC-01 to UC-05), incorporating both «include» (session verification, real-time tally) and «extend» (spoiler reveal, milestone badge award) relationships along with actor generalization.



### 3. Use-Case Flow Specification: UC-04 (Vote in Monthly Book Selection Poll)

|                                                                                                                                                                               |                                                           |                                                                                                                                                                                                                         |                                      |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------|
| <b>Use Case ID:</b>                                                                                                                                                           | UC-04                                                     | <b>Use Case Name:</b>                                                                                                                                                                                                   | Vote in Monthly Book Selection Poll  |
| <b>Primary Actor:</b>                                                                                                                                                         | Club Member                                               | <b>Secondary Actors:</b>                                                                                                                                                                                                | Auth Service, Real-Time Tally Engine |
| <b>Stereotypes:</b>                                                                                                                                                           | «include» Session Verification, «include» Real-Time Tally | <b>Priority:</b>                                                                                                                                                                                                        | High (Core Democratic Feature)       |
| <b>Preconditions:</b><br>1. User is authenticated with active session.<br>2. Monthly book selection poll is active and open.<br>3. Member has not yet voted in current cycle. |                                                           | <b>Postconditions:</b><br>1. Member's single vote is recorded and permanently bound.<br>2. Live aggregate tallies update across all active user sessions.<br>3. Member UI transitions to locked read-only results view. |                                      |

#### Main Success Scenario (MSS):

1. **Navigate to Poll:** Member navigates to the 'Monthly Book Poll' section on dashboard.
2. **Retrieve Details:** System retrieves active poll candidates, synopsis, and checks voting status.
3. **Display Nominees:** System renders nominated books with selection radio options.
4. **Select Candidate:** Member selects their preferred book title.
5. **Submit Vote:** Member clicks 'Submit Vote'.
6. **Confirm Modal:** System displays confirmation modal ('Confirm vote for [Book Title]?').
7. **Acknowledge:** Member confirms selection.
8. **Validate & Persist:** System checks user session and performs atomic write transaction.
9. **Update Tallies:** System updates Redis real-time count and emits websocket broadcast.
10. **Render Confirmation:** System displays success banner and locks member voting controls.
11. **Termination:** Use case completes successfully.

#### Alternate / Exceptional Flows:

- 3a. Already Voted in Cycle:** System identifies prior vote → Displays live percentage chart & disables vote button → Terminates.
- 5a. Empty Selection Submitted:** Member clicks submit without choice → System highlights options in red & prompts selection → Resumes Step 5.
- 8a. Duplicate Submission Collision:** Database unique key constraint flags collision → System rolls back and returns 409 Conflict → Displays alert → Terminates.
- 8b. Poll Expiration During Voting:** System detects poll deadline passed → Informs user poll has closed → Redirects to final results → Terminates.